

coding challenge 7

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Question 1

```
library(tidyverse)

## Warning: package 'tidyverse' was built under R version 4.5.2
## Warning: package 'ggplot2' was built under R version 4.5.2
## Warning: package 'tibble' was built under R version 4.5.2
## Warning: package 'tidyr' was built under R version 4.5.2
## Warning: package 'readr' was built under R version 4.5.2
## Warning: package 'purrr' was built under R version 4.5.2
## Warning: package 'dplyr' was built under R version 4.5.2
## Warning: package 'stringr' was built under R version 4.5.2
## Warning: package 'forcats' was built under R version 4.5.2
## Warning: package 'lubridate' was built under R version 4.5.2

## — Attaching core tidyverse packages ————— tidyverse
2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.6
## ✓ forcats   1.0.1      ✓ stringr    1.6.0
## ✓ ggplot2    4.0.2      ✓ tibble     3.3.1
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.2
## ✓ purrr     1.2.1
```

```
## — Conflicts —————
tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag() masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force
all conflicts to become errors
```

```
library(lme4)
```

```
## Warning: package 'lme4' was built under R version 4.5.2
```

```
## Loading required package: Matrix
```

```
##
```

```
## Attaching package: 'Matrix'
```

```
##
```

```
## The following objects are masked from 'package:tidyr':
```

```
##
```

```
## expand, pack, unpack
```

```
library(emmeans)
```

```
## Warning: package 'emmeans' was built under R version 4.5.3
```

```
## Welcome to emmeans.
```

```
## Caution: You lose important information if you filter this package's
results.
```

```
## See '? untidy'
```

```
library(multcomp)
```

```
## Warning: package 'multcomp' was built under R version 4.5.2
```

```
## Loading required package: mvtnorm
```

```
## Warning: package 'mvtnorm' was built under R version 4.5.3
```

```
## Loading required package: survival
```

```
## Loading required package: TH.data
```

```
## Warning: package 'TH.data' was built under R version 4.5.2
```

```
## Loading required package: MASS
```

```
##
```

```
## Attaching package: 'MASS'
```

```
##
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
## select
```

```
##
```

```
##
```

```
## Attaching package: 'TH.data'
```

```
##
```

```
## The following object is masked from 'package:MASS':
##
##      geyser

library(multcompView)

## Warning: package 'multcompView' was built under R version 4.5.3

Plant <- read.csv("PlantEmergence.csv")
Plant$Treatment <- as.factor(Plant$Treatment)
Plant$DaysAfterPlanting <- as.factor(Plant$DaysAfterPlanting)
Plant$Rep <- as.factor(Plant$Rep)
```

Question 2

```
model <- lm(Emergence~Treatment*DaysAfterPlanting, data = Plant)
summary(model)
```

```
##
## Call:
## lm(formula = Emergence ~ Treatment * DaysAfterPlanting, data = Plant)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-21.250	-6.062	-0.875	6.750	21.875

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.823e+02	5.324e+00	34.229	<2e-16	***
Treatment2	-1.365e+02	7.530e+00	-18.128	<2e-16	***
Treatment3	1.112e+01	7.530e+00	1.477	0.142	
Treatment4	2.500e+00	7.530e+00	0.332	0.741	
Treatment5	8.750e+00	7.530e+00	1.162	0.248	
Treatment6	7.000e+00	7.530e+00	0.930	0.355	
Treatment7	-1.250e-01	7.530e+00	-0.017	0.987	
Treatment8	9.125e+00	7.530e+00	1.212	0.228	
Treatment9	2.375e+00	7.530e+00	0.315	0.753	
DaysAfterPlanting14	1.000e+01	7.530e+00	1.328	0.187	
DaysAfterPlanting21	1.062e+01	7.530e+00	1.411	0.161	
DaysAfterPlanting28	1.100e+01	7.530e+00	1.461	0.147	
Treatment2:DaysAfterPlanting14	1.625e+00	1.065e+01	0.153	0.879	
Treatment3:DaysAfterPlanting14	-2.625e+00	1.065e+01	-0.247	0.806	
Treatment4:DaysAfterPlanting14	-6.250e-01	1.065e+01	-0.059	0.953	
Treatment5:DaysAfterPlanting14	2.500e+00	1.065e+01	0.235	0.815	
Treatment6:DaysAfterPlanting14	1.000e+00	1.065e+01	0.094	0.925	
Treatment7:DaysAfterPlanting14	-2.500e+00	1.065e+01	-0.235	0.815	
Treatment8:DaysAfterPlanting14	-2.500e+00	1.065e+01	-0.235	0.815	
Treatment9:DaysAfterPlanting14	6.250e-01	1.065e+01	0.059	0.953	
Treatment2:DaysAfterPlanting21	3.500e+00	1.065e+01	0.329	0.743	
Treatment3:DaysAfterPlanting21	-1.000e+00	1.065e+01	-0.094	0.925	

```
## Treatment4:DaysAfterPlanting21 1.500e+00 1.065e+01 0.141 0.888
## Treatment5:DaysAfterPlanting21 2.875e+00 1.065e+01 0.270 0.788
## Treatment6:DaysAfterPlanting21 4.125e+00 1.065e+01 0.387 0.699
## Treatment7:DaysAfterPlanting21 -2.125e+00 1.065e+01 -0.200 0.842
## Treatment8:DaysAfterPlanting21 -1.500e+00 1.065e+01 -0.141 0.888
## Treatment9:DaysAfterPlanting21 -1.250e+00 1.065e+01 -0.117 0.907
## Treatment2:DaysAfterPlanting28 2.750e+00 1.065e+01 0.258 0.797
## Treatment3:DaysAfterPlanting28 -1.875e+00 1.065e+01 -0.176 0.861
## Treatment4:DaysAfterPlanting28 3.264e-13 1.065e+01 0.000 1.000
## Treatment5:DaysAfterPlanting28 2.500e+00 1.065e+01 0.235 0.815
## Treatment6:DaysAfterPlanting28 2.125e+00 1.065e+01 0.200 0.842
## Treatment7:DaysAfterPlanting28 -3.625e+00 1.065e+01 -0.340 0.734
## Treatment8:DaysAfterPlanting28 -1.500e+00 1.065e+01 -0.141 0.888
## Treatment9:DaysAfterPlanting28 -8.750e-01 1.065e+01 -0.082 0.935
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.65 on 108 degrees of freedom
## Multiple R-squared:  0.9585, Adjusted R-squared:  0.945
## F-statistic: 71.21 on 35 and 108 DF, p-value: < 2.2e-16
```

```
anova(model)
```

```
## Analysis of Variance Table
##
## Response: Emergence
##
##           Df Sum Sq Mean Sq  F value    Pr(>F)
## Treatment      8 279366    34921 307.9516 < 2.2e-16 ***
## DaysAfterPlanting 3   3116     1039   9.1603 1.877e-05 ***
## Treatment:DaysAfterPlanting 24    142         6  0.0522      1
## Residuals    108  12247      113
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 3

- No, we do not need to fit the interaction term.

```
model_simple <- lm(Emergence ~ Treatment + DaysAfterPlanting, data = Plant)
summary(model_simple)
```

```
##
## Call:
## lm(formula = Emergence ~ Treatment + DaysAfterPlanting, data = Plant)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.1632  -6.1536  -0.8542   6.1823  21.3958
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)          182.163      2.797  65.136 < 2e-16 ***
## Treatment2          -134.531      3.425 -39.277 < 2e-16 ***
## Treatment3           9.750      3.425   2.847 0.00513 **
## Treatment4           2.719      3.425   0.794 0.42876
## Treatment5          10.719      3.425   3.129 0.00216 **
## Treatment6           8.812      3.425   2.573 0.01119 *
## Treatment7          -2.188      3.425  -0.639 0.52416
## Treatment8           7.750      3.425   2.263 0.02529 *
## Treatment9           2.000      3.425   0.584 0.56028
## DaysAfterPlanting14   9.722      2.283   4.258 3.89e-05 ***
## DaysAfterPlanting21  11.306      2.283   4.951 2.21e-06 ***
## DaysAfterPlanting28  10.944      2.283   4.793 4.36e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.688 on 132 degrees of freedom
## Multiple R-squared:  0.958, Adjusted R-squared:  0.9545
## F-statistic: 273.6 on 11 and 132 DF, p-value: < 2.2e-16

anova(model_simple)

## Analysis of Variance Table
##
## Response: Emergence
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Treatment      8 279366    34921 372.070 < 2.2e-16 ***
## DaysAfterPlanting 3   3116     1039  11.068 1.575e-06 ***
## Residuals     132  12389         94
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Intercept (182.163),Treatment 2 Coefficient (-134.531)

Question 4

```
lsmeans <- emmeans(model_simple, ~Treatment) #LeastSquared means
Results_lsmeans <- cld(lsmeans, alpha = 0.05, reversed = TRUE, details = TRUE)
```

Interpret:The least square means (LSMeans) and Tukey grouping show that Treatment 2 is significantly different from all other treatments.

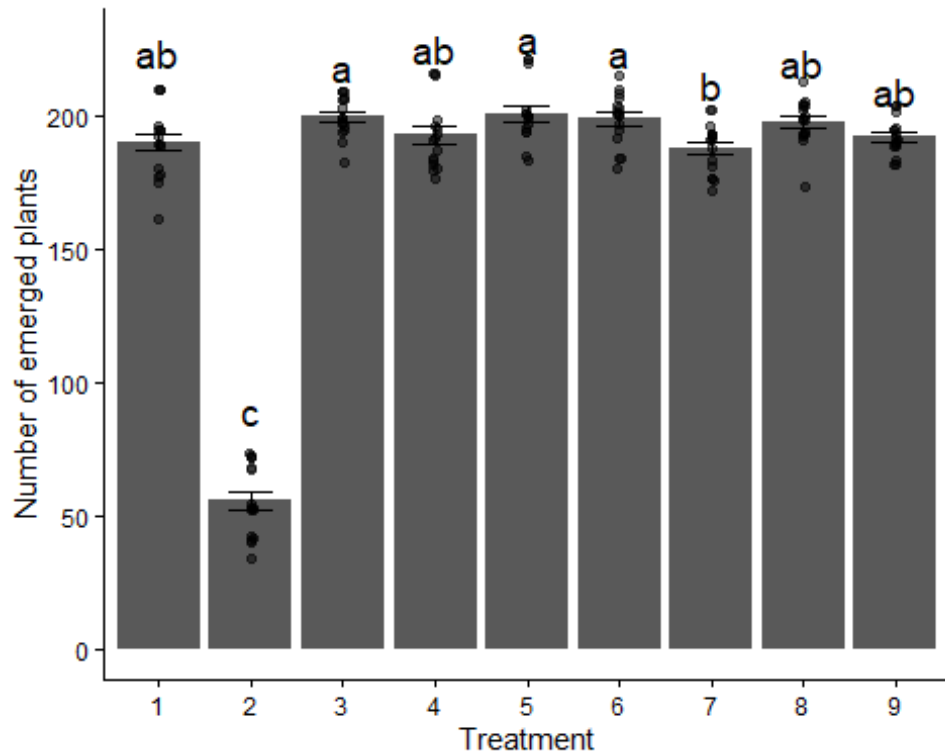
Question 5

```
plot_cldbars_onefactor <- function(lm_model, factor) {
  data <- lm_model$model
  variables <- colnames(lm_model$model)
  dependent_var <- variables[1]
  independent_var <- variables[2:length(variables)]
```

```

lsmeans <- emmeans(lm_model, as.formula(paste("~", factor))) # estimate
lsmeans
Results_lsmeans <- cld(lsmeans, alpha = 0.05, reversed = TRUE, details =
TRUE, Letters = letters) # contrast with Tukey adjustment by default.
# Extracting the Letters for the bars
sig.diff.letters <- data.frame(Results_lsmeans$emmeans[,1],
str_trim(Results_lsmeans$emmeans[,7]))
colnames(sig.diff.letters) <- c(factor, "Letters")
# for plotting with Letters from significance test
ave_stand2 <- lm_model$model %>%
group_by(!sym(factor)) %>%
dplyr::summarize(
ave.emerge = mean(.data[[dependent_var]], na.rm = TRUE),
se = sd(.data[[dependent_var]]) / sqrt(n())
) %>%
left_join(sig.diff.letters, by = factor) %>%
mutate(letter_position = ave.emerge + 10 * se)
plot <- ggplot(data, aes(x = !! sym(factor), y = !! sym(dependent_var))) +
stat_summary(fun = mean, geom = "bar") +
stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.5) +
ylab("Number of emerged plants") +
geom_jitter(width = 0.02, alpha = 0.5) +
geom_text(data = ave_stand2, aes(label = Letters, y = letter_position),
size = 5) +
xlab(as.character(factor)) +
theme_classic()
return(plot)
}
plot_cldbars_onefactor(model_simple, "Treatment")

```



The intercept represents the mean emergence for the reference treatment at the reference day level, and the coefficient for Treatment 2 shows how much emergence differs from the reference treatment after adjusting for DaysAfterPlanting.

Question 6

<https://github.com/Hikmat-au/Assignments-repository/tree/main/Coding%20challenge%207>